

Securing Devices -- Guided Notes ANSWER KEY

Unit tp-6.10 | CompTIA Network+ N10-009 Obj. FC0-U71 6.2 | N10-008 FC0-U71 6.2

For instructor use / distribute after students complete the notes.

Question 1

Endpoint security refers to protecting _____ devices (laptops, phones, tablets, desktops) from threats. Each endpoint is a potential _____ into the network. Key endpoint controls include _____ software, _____ encryption, _____ management, and device _____ policies.

Answer: End-user; entry point (attack vector); antivirus; full-disk; patch; hardening.

Question 2

Full disk encryption (FDE) protects data on a device if it is lost or _____. BitLocker is the FDE solution built into Windows _____. FileVault is the FDE solution built into _____. When FDE is enabled, all data on the drive is _____ and requires an authentication step at boot to unlock.

Answer: Stolen; Pro/Enterprise; macOS; encrypted.

Question 3

Patch management is the process of applying _____ updates to operating systems and applications. Unpatched vulnerabilities are the leading cause of successful _____. Organizations should apply critical patches within _____ days of release. _____ testing or patch testing in a non-production environment before deploying reduces risk of breaking systems.

Answer: Security; attacks (breaches); 30 (14 or 72 hours for critical); staged.

Question 4

Antivirus (AV) software detects known malware using _____ -based detection (comparing files against a database of known malware signatures). Modern _____ (Endpoint Detection and Response) tools use behavior-based detection to catch _____ malware that has no known signature. Both types should generate _____ when threats are found.

Answer: Signature; EDR; zero-day (novel); alerts.

Question 5

A host-based _____ (HFW) controls which network connections a device can make and accept. It blocks _____ access attempts from the network. The Windows Defender Firewall is a built-in _____ for Windows. Rules specify allowed and denied _____ (80/443 for web, 22 for SSH) and source _____.

Answer: Firewall; unauthorized (inbound); HFW; ports; IP addresses.

Question 6

MDM (Mobile Device Management) allows organizations to _____, configure, and _____ employee mobile devices. MDM can enforce _____ requirements, push _____ updates, and remotely _____ a lost or stolen device. BYOD (Bring Your Own Device) policies require MDM

enrollment to access corporate resources.

Answer: Enroll; monitor; password; configuration/security; wipe.

Question 7

Device hardening reduces the _____ surface by removing or disabling unnecessary services, software, and features. Steps include: disable unused _____ (Bluetooth, Wi-Fi, IR), remove unneeded _____ (games, media players on servers), change default _____, and enable _____ audit logging.

Answer: Attack; interfaces; software (applications); passwords; system.

Question 8 -- Real World Application

REAL WORLD: A healthcare worker accidentally leaves their unencrypted company laptop at a coffee shop. The laptop contains a local cache of 300 patient appointment records including names, DOBs, and phone numbers. The laptop is not recovered. Describe the regulatory obligations the healthcare organization now has and explain what two device security controls would have prevented this from being a reportable breach.

Answer: Regulatory obligations: this is a HIPAA breach. Because the PHI (protected health information) was unencrypted and on an unrecovered device, HHS must be notified. If 300+ individuals are affected, notification to HHS must occur within 60 days of year end (or 60 days of discovery if 500+ individuals). Affected patients must receive written breach notification letters. The state attorney general may also require notification under state breach notification law. Control 1 -- Full disk encryption (BitLocker): if the laptop's drive was BitLocker-encrypted with a strong recovery key managed by the organization, the thief cannot access any data without the encryption key -- even by removing the drive and connecting to another machine. HHS guidance explicitly states that encrypted lost/stolen devices are NOT reportable breaches, because the data is cryptographically protected. Control 2 -- MDM with remote wipe: enroll all laptops in an MDM solution. If the device connects to any network, the MDM can issue a remote wipe, erasing all data. This limits the exposure window and demonstrates to regulators that the organization took rapid mitigation steps.